

Negative effect of a low-carbohydrate, high-protein, high-fat diet on small peripheral artery reactivity in patients with increased cardiovascular ...



Low-carbohydrate diets have become increasingly popular for weight loss. Although they may improve some metabolic markers, particularly in type 2 diabetes mellitus (T2D) or the metabolic syndrome (MS), their net effect on arterial wall function remains unclear. The objective was to evaluate the relation between dietary macronutrient composition and the small artery reactive hyperaemia index (saRHI), a marker of small artery endothelial function, in a cohort of patients at increased cardiovascular (CV) risk. The present cross-sectional study included 247 patients. Diet was evaluated by a 3-d food-intake register and reduced to a novel low-carbohydrate diet score (LCDS). Physical examination, demographic, biochemical and anthropometry parameters were recorded, and the saRHI was measured in each patient. Individuals in the lowest LCDS quartile (Q1, 45 % carbohydrate; 20 % protein; 32 % fat) had higher saRHI values than those in the top quartile (Q4, 29 % carbohydrate, 24 % protein, 40 % fat; 1.66 (sd 0.41) v. 1.52 (sd 0.22), $P=0.037$). These results were particularly strong in patients with the MS (Q1 = 1.82 (sd 0.32) v. Q4 = 1.61 (sd 0.27); $P=0.021$) and T2D (Q1 = 1.78 (sd 0.31) v. Q4 = 1.62 (sd 0.35); $P=0.011$). Multivariate analysis demonstrated that individuals in the highest LCDS quartile had a significantly negative coefficient of saRHI, which was independent of confounders (OR -0.85; 95 % CI 0.19, 0.92; $P=0.031$). These findings suggest that a dietary pattern characterised by a low amount of carbohydrate, but high amounts of protein and fat, is associated with a poorer small artery vascular reactivity in patients with increased CV risk.